OPERATING SYSTEMS CONCEPT – LAB

EXP – 05: SCHEDULING ALGORITHMS – FCFS, SJF, RR AND PRIORITY

FCFS ALGORITHM

Activities   Terminal   Apr 20 17:43

snoordeen@localhost:~/os-lab/scheduling — /usr/bin/vim fcfs.c

```
snoordeen@localhost:~/os-lab/scheduling$ cat fcfs.c
#include <stdio.h>
int main(){
        int n;
        printf("Enter number of processes: ");
        scanf("%d", &n);
        int at[n], bt[n], ct[n], tat[n], wt[n];
        for(int i=0; i<n; i++){
                printf("Enter Arrival time and Burst time for P%d :",i+1);
                scanf("%d %d", &at[i], &bt[i]);
        }
        float total_tat=0, total_wt=0;
        ct[0]=at[0]+bt[0];
        for(int i=1; i<n; i++){
                if(ct[i-1]>at[i])
                        ct[i]=ct[i-1]+bt[i];
                else
                        ct[i]=at[i-1]+bt[i];
                }
        }
        for(int i=0; i<n; i++){
                tat[i]=ct[i]-at[i];
                wt[i]=tat[i]-bt[i];
                total_tat+=tat[i];
                total_wt+=wt[i];
        }
        printf("\nAverage Turnaround Time: %.2f", total_tat/n);
        printf("\nAverage Waiting Time: %.2f\n", total_wt/n);
        return 0;
}
```

Activities   Terminal   Apr 20 17:45

snoordeen@localhost:~/os-lab/scheduling

```
snoordeen@localhost:~/os-lab/scheduling$ vi fcfs.c
snoordeen@localhost:~/os-lab/scheduling$ gcc fcfs.c -o fcfs
snoordeen@localhost:~/os-lab/scheduling$ ./fcfs
Enter number of processes: 4
Enter Arrival time and Burst time for P1: 0 2
Enter Arrival time and Burst time for P2: 1 3
Enter Arrival time and Burst time for P3: 2 6
Enter Arrival time and Burst time for P4: 3 4
Average Turnaround Time: 8.50
Average Waiting Time: 6.75
snoordeen@localhost:~/os-lab/scheduling$
```

SJF ALGORITHM

Activities Terminal Apr 20 19:46

snoordeen@localhost:~/os-lab/scheduling — /usr/bin/vim sjf.c

```c
#include <stdio.h>
int main(){
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n], bt[n], ct[n], tat[n], wt[n];
    int done[n], i, j, min, idx;
    for(int i=0; i<n; i++){
        printf("Enter Arrival Time and Burst Time for P%d :", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        done[i]=0;
    }
    int completed=0;
    int completionTime=0;
    float avgTat=0, avgWt=0;
    while(completed<n){
        int min=99999;
        for(int j=0; j<n; j++){
            if(done[j]==0 && at[j]<=completionTime && bt[j]<min){
                min=bt[j];
                idx=j;
            }
        }
        if(idx!=-1){
            if(completionTime<at[idx]){
                completionTime=at[idx];
            }
            completionTime+=bt[idx];
            ct[idx]=completionTime;
            tat[idx]=ct[idx]-at[idx];
            wt[idx]=tat[idx]-bt[idx];
            done[idx]=1;
            completed++;
        } else {
            completionTime++;
        }
    }
    for(int i=0; i<n; i++){
        avgTat+=tat[i];
    }   avgWt+=wt[i];
    printf("Average Turnaround Time: %.2f\n", avgTat/n);
    printf("Average Waiting Time: %.2f\n", avgWt/n);
    return 0;
}
                                                    43,8-1        Bot
```

SJF ALGORITHM OUTPUT

Activities Terminal Apr 20 19:48

snoordeen@localhost:~/os-lab/scheduling

```
snoordeen@localhost:~/os-lab/scheduling$ vi sjf.c
snoordeen@localhost:~/os-lab/scheduling$ gcc sjf.c -o sjf
snoordeen@localhost:~/os-lab/scheduling$ ./sjf
Enter number of processes: 4
Enter Arrival Time and Burst Time for P1 : 0 7
Enter Arrival Time and Burst Time for P2 : 1 3
Enter Arrival Time and Burst Time for P3 : 2 6
Enter Arrival Time and Burst Time for P4 : 3 4
Average Turnaround Time: 8.75
Average Waiting Time: 3.00
snoordeen@localhost:~/os-lab/scheduling$
```